Governor House AI Summer Camp — Faculty Selection Challenge

Game link = https://summer-camp-game.vercel.app/
Chat link = https://chat.deepseek.com/share/bzjsnldoii6xrxcrmb
Concept taught = “# Pixel's AI Academy – Game Concept Documentation

1. Introduction

Pixel's AI Academy is an interactive, story-driven HTML game designed for teenagers (ages 13–18) to learn core AI concepts through hands-on problem-solving. The game wraps each concept in a friendly, futuristic park setting where players assist Pixel – a cheerful robot who has just graduated from AI school – in running his "Help Booth." Visitors arrive with various requests, and the player must choose the right AI “tool” to help them. The game avoids traditional quizzes and instead teaches through **consequence-based feedback**, animated character reactions, and a gradual progression through six scenarios per concept.

The first and fully playable concept is **The Three Retrieval Modes** (Memory, Search, Research). Three additional concepts – *Supervised vs Unsupervised*, *Reinforcement Learning*, and *Neural Networks* – are also included as separate levels, each with its own set of tools and visitors.

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2. The Core Concept: The Three Retrieval Modes

At the heart of the first level is the idea that an AI system can access information in three fundamentally different ways:

- **🧠 Memory** – using static, unchanging knowledge stored in the system (like a database of facts, historical events, or basic science). This is suitable for questions that have a single, stable answer.

- **🔍 Search** – retrieving up-to-the-minute information from the web or a live data source. This is ideal for questions about current events, weather, trending topics, or anything that changes over time.

- **📚 Research** – performing a deep, analytical investigation that combines multiple sources, synthesises information, and draws conclusions. This is needed for complex, open-ended problems that require understanding systems or building new knowledge.

The key skill the game teaches is **knowing which mode to apply in a given situation** – a crucial ability for both human and AI decision-making.

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3. Gameplay Overview

The game is structured as a series of **visitor encounters**. Each visitor presents a problem in a speech bubble. The player must select one of three tools (the three retrieval modes) to help them. The game then provides an immediate consequence:

- **Correct choice** → Pixel gives a perfect, relevant answer; the visitor reacts with joy (dances, emoji changes) and a happy sound plays.
- **Incorrect choice** → Pixel gives an outdated, shallow, or inappropriate answer; the visitor looks confused or disappointed (shakes head, sad emoji) and a buzzer sound plays.

There is **no score**, **no right/wrong popup**, and **no percentage**. The player learns solely by observing the characters' reactions and the content of Pixel's responses. After six visitors, the level concludes with a "Graduation Party" animation, celebrating the player's learning journey.

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4. Characters and Story

- **Pixel** – A bright, enthusiastic robot who runs the Help Booth. He is friendly, slightly clumsy when he makes mistakes, and eager to learn alongside the player. Pixel provides the in-game hints and reacts to every choice with a unique emoji and text.

- **Visitors** – Each visitor has a distinct name, emoji, and personality. They arrive with a realistic problem that matches one of the three retrieval modes. Their reactions (happy dance, confused shake) are the primary feedback mechanism.

- **Story context** – Pixel has just graduated from AI school and wants to prove he can help everyone. The player becomes Pixel's assistant, helping him choose the right tool for each visitor. The game progresses through six visitors, and after helping them all, Pixel "graduates" and earns a "Top Assistant Badge."

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5. The Learning Cycle

The game employs a **learning-by-doing** approach with the following cycle:

1. **Encounter** – A visitor arrives with a problem.
2. **Decision** – The player selects one of the three tools.
3. **Consequence** – Pixel responds with an answer; the visitor and Pixel react emotionally.

4. **Reflection** – The player observes the outcome and understands why the tool was (or was not) appropriate.
5. **Repetition** – The cycle repeats with a new visitor, reinforcing the concept through varied examples.

Importantly, after an incorrect choice, the player is given the opportunity to try again with the same visitor (the visitor does not leave until helped correctly). This eliminates frustration and encourages experimentation.

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6. Additional Concepts (Other Levels)

Beyond the Three Retrieval Modes, *Pixel's AI Academy* includes three additional playable levels, each teaching a distinct AI concept:

- **📊 Supervised vs Unsupervised** – Players learn to distinguish between learning from labeled data (supervised) and discovering hidden patterns without labels (unsupervised). Tools: *Labeled Data*, *Pattern Finder*, and *Random Guess*.

- **🎮 Reinforcement Learning** – Players practice giving feedback (reward, punishment, or ignore) to train an agent to perform desired behaviours. Tools: *Reward*, *Punish*, *Ignore*.

- **🧩 Neural Networks** – Players explore the building blocks of deep learning: adding layers, adjusting weights, and choosing activation functions. Tools: *Add Layers*, *Adjust Weights*, *Activation*.

Each level has its own set of visitors, tools, and a dedicated concept guide accessible via the **📖 Concept** button.

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7. Visual and Audio Design

The game is designed to be **bright, colourful, and cartoon-like**, appealing to a teenage audience:

- **Backgrounds** – A sunny park scene with trees, clouds, flowers, and a spinning sun. Day, Night, and Gaming themes are available to change the mood.
- **Characters** – Expressive emoji-based characters (🤖 for Pixel, and various emojis for visitors) that animate (bounce, dance, shake) to convey emotions.
- **UI Elements** – Large, rounded buttons with emojis and clear labels, making them easy to tap on mobile devices.

- **Sound Effects** – Simple Web Audio API sounds for clicks, correct choices (happy chime), and incorrect choices (funny buzzer). A mute/unmute button is provided.
- **Animations** – Smooth transitions, floating clouds, rotating sun, and confetti celebrations for graduation.

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8. Educational Value

Pixel’s AI Academy is designed with pedagogical principles in mind:

- **Active Learning** – Players are constantly making decisions and observing outcomes, which strengthens retention.
- **Immediate Feedback** – Consequences are delivered instantly, allowing players to connect actions to results.
- **Low-Stakes Environment** – No penalties for mistakes; players can try again, reducing anxiety.
- **Scaffolding** – A hint system (💡 Hint button) provides gentle guidance when needed.
- **Transferable Skills** – The ability to choose the right information retrieval method is a valuable real-world skill, applicable beyond AI.
- **Intrinsic Motivation** – The narrative of helping Pixel and watching his progress encourages continued play without external rewards.

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9. How to Play

1. **Start the Game** – From the home screen, click any playable concept card (e.g., "Three Retrieval Modes").
2. **Read the Visitor's Problem** – A speech bubble appears above the visitor.
3. **Choose a Tool** – Click one of the three large tool buttons at the bottom of the screen.
4. **Observe the Reaction** – Pixel will respond, and the visitor’s expression will change. Listen to the sound feedback.
5. **Repeat** – Continue helping visitors until all six are satisfied.
6. **Celebrate** – Watch the graduation animation and enjoy the confetti!
7. **Explore Other Concepts** – Return to the home screen (🏠 Home) to try the other levels.

Keyboard Shortcuts (for desktop):
- `1`, `2`, `3` – select the first, second, or third tool.
- `H` – show a hint.
- `M` – mute/unmute.
- `C` – open the concept guide.
- `Esc` – return to home (from game screen) or close the concept modal.

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10. Final Remarks

Pixel’s AI Academy transforms abstract AI concepts into tangible, memorable experiences. By blending storytelling, interactive problem-solving, and playful visuals, the game makes learning both effective and enjoyable. Whether you're a student, teacher, or lifelong learner, stepping into Pixel’s booth is a delightful way to understand how AI retrieves and processes information.

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